YINHONG LIU

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EDUCATION EXPERIENCE

University of Cambridge, Language Technology Lab; PhD **2021 - 2026**

- Supervisor: Prof. Nigel Collier
- Research area: Large Language Model (LLM) Evaluation and Alignment. Text and Multi-modal Evaluation Techniques. Controllable Text Generation. Data Synthesis for Alignment and Down-stream applications. Advanced Dialog Systems. Discourse Structure Analysis for Long-form Text.

University of Cambridge, Computer Lab; Mphil **2020 - 2021**

- Subject: **Advanced Computer Science, in Natural Language Processing**.
- Grade: **Distinction**.
- Research area: Learning Functional Distributional Semantics from Multi-modal data.

University of Cambridge, Engineering Department; BA, MEng **2014 - 2018**

- Subject: **Information Engineering and Computer Science**.
- Grade in Master of Engineering: **Merit degree** with honours.
- Grade in Bachelor of Arts: **1st Class**.

WORK, RESEARCH EXPERIENCE

Tencent-HY; Shenzhen. *QingYun Plan* **Nov. 2026 - Present**

· Working on full-duplex interaction agentic models.

Yinwang (Huawei); Shenzhen. *Genius Youth/TopMinds* **Feb. 2026 - Nov. 2026**

· Working on full-duplex omni-modal interaction models and agent frameworks for in-vehicle cabin scenarios.

· Tech Lead; Designed the interaction model implementation method; Developed data curation, post-training, inference framework, and designed training pipeline.

Amazon AWS AI; Seattle. *Applied Scientist Intern* **Oct. 2024 - Feb. 2025**

· Develop Multi-Modal Summarization meta-Evaluation benchmark and evaluation methods.

Mircrosoft Research; Cambridge. *Research Intern* **Jun. 2024 - Sep. 2024**

· Developed an advanced Automatic Prompt Optimization pipeline to enhance Model Migration efficiency and improve Model Routing for Microsoft Copilot teams.

AI/ML, Apple; Cambridge. *Research Intern* **Jun. 2023 - Oct. 2023**

· Developed an automatic Task-Oriented Dialog generation pipeline, incorporating advanced analysis of future virtual assistant response styles and simulating complex apps interactions. This work benefits both internal Apple Intelligence development and external NLP community.

Gambit Research; London. *Full-time Quant-analyst* **Oct. 2018 - Jul. 2020**

· Working on various statistical football teams & games modelings, market price & volume predictions, strategies developing and back-testing.

SELECTED PUBLICATIONS

Recursive Self-Improvement & Agents:

• *Data-Centric Recursive Improvement for Foundation Models: A Survey.* Under Review. Co-author.

• *From Trainee to Trainer: LLM-Designed Training Environment for RL with Multi-Agent Reasoning.* Under Review. Advisor.

• *Action-Space Shaping for LLM Agents: Measuring and Mitigating Tool-Schema Bias.* Under Review. First author.

• *Trajectory-Level Speculation with Semantic Verification for Tool-Using Agents.* Under Review. Advisor.

LLM Evaluation & LLM-as-a-Judge:

• *Aligning with Human Judgement: The Role of Pairwise Preference in Large Language Model Evaluators.* COLM 2024. First author.

• *MDSEval: A Meta-Evaluation Benchmark for Multimodal Dialogue Summarization.* EMNLP 2025. First author.

• *Unlocking Structure Measuring: Introducing PDD, an Automatic Metric for Positional Discourse Coherence.* NAACL 2024. First author.

Preference Alignment & Reasoning:

• *Aligning with Logic: Measuring, Evaluating and Improving Logical Consistency in Large Language Models.* ICML 2025, Spotlight. First author.

• *Operadic consistency: a label-free signal for compositional reasoning failures in LLMs.* NeurIPS 2026. Co-author.

• *When Personalization Meets Reality: A Multi-Faceted Analysis of Personalized Preference Learning.* EMNLP 2025. Co-author.

• *Improving Preference Extraction In LLMs By Identifying Latent Knowledge Through Classifying Probes.* ACL 2025. Advisor.

• *Fairer Preferences Elicit Improved Human-Aligned Large Language Model Judgments.* EMNLP 2024. Co-author.

Agentic Environment & Data Synthesis:

• *EnvFactory: Scaling Tool-Use Agents via Executable Environments Synthesis and Robust RL.* Under Review 2026, Co-author.

• *Dial HEALTHDIAL for Advice: A Multilingual and Multi-Parallel Spoken Dialogue Dataset for Evidence-Based Public Health Communication.* ACL 2026, First author.

• *TOAD: Task-Oriented Automatic Dialog generation pipeline with diverse system response style.* ACL 2024. First author.

Others:

• *From Chains to Graphs: Self-Structured Reasoning for General-Domain LLMs.* AACL-IJCNLP 2026. Co-author.

• *GraphCheck: Breaking Long-Term Text Barriers with Extracted Knowledge Graph-Powered Fact-Checking.* ACL 2025. Co-author.

• *FASTFACT: Faster, Stronger Long-Form Factuality Evaluations in LLMs.* EMNLP 2025. Co-author.

• *Prompt Compression for Large Language Models: A Survey.* NAACL 2025. First author.

PRIZE & EXTRA-CURRICULUM ACTIVITIES

Best Reviewer Award, EMNLP 2023, Natural Language Generation track.

PhD scholarship: Cambridge International Scholarship by Cambridge Trust.

Academic scholarship: Caldwell Scholarship by Corpus Christi College, University of Cambridge